$\boxed{1.}$ For N massless Weyl fermions ψ_j described by free Lagrangian density

$$\mathcal{L} = i\psi_j^\dagger \sigma^\mu \partial_\mu \psi_j$$

is the transformation of the fields

$$\psi_j \to M_{ji}\psi_i$$

globally symmetric iff

$$M^*_{kj}M_{ji} = \delta_{ki} \qquad \Leftrightarrow \qquad M^\dagger M = \mathbb{1}.$$

The continuous global symmetry group is therefore $\mathrm{U}(N)$.

$\boxed{2.}$

(i) In the case of N Weyl fields with a common mass

$$\mathcal{L} = i\psi_j^\dagger \sigma^\mu \partial_\mu \psi_j - \frac{1}{2}m(\psi_j^{\mathrm{T}}\epsilon\psi_j + \psi_j^\dagger \epsilon \psi_j^*),$$

must matrix M satisfy the following conditions

$$M^*_{kj}M_{ji} = \delta_{ki}, \qquad M_{kj}M_{ji} = \delta_{ki} \qquad \text{and} \qquad M^*_{kj}M^*_{ji} = \delta_{ki},$$

which are all satisfied iff $M^{\mathrm{T}}M = \mathbb{1}$ and $M \in \mathbb{R}^{\mathrm{N}\times\mathrm{N}}$. If we are interested only in the continuous global symmetry connected to identity, then the searched for symmetry group is $\mathrm{SO}(N)$, if we also consider the $\mathbb{Z}_2$ discrete symmetry, then the symmetry group is $\mathrm{O}(N)$

(ii) In the case of N massless Majorana fields we can rewrite the Lagrangian density

$$\mathcal{L} = \frac{i}{2}\Psi_j^{\mathrm{T}} C\gamma^\mu \partial_\mu \Psi_j$$

using the relation of the components of a Majorana field

$$\Psi_j = \begin{pmatrix} \psi_j \\ i\sigma^2\psi_j^* \end{pmatrix}$$

obtaining

$$\mathcal{L} = \frac{i}{2}\begin{pmatrix} \psi_j^{\mathrm{T}} & i(\sigma^2\psi_j^*)^{\mathrm{T}} \end{pmatrix} \begin{pmatrix} -i\sigma^2 & 0 \\ 0 & i\sigma^2 \end{pmatrix} \begin{pmatrix} 0 & \bar{\sigma}^\mu \\ \sigma^\mu & 0 \end{pmatrix} \partial_\mu \begin{pmatrix} \psi_j \\ i\sigma^2\psi_j^* \end{pmatrix}$$

$$= \frac{i}{2}\psi_j^{\mathrm{T}}(-i\sigma^2)\bar{\sigma}^\mu\partial_\mu i\sigma^2\psi_j^* + \frac{i}{2}(i\sigma^2\psi_j^*)^{\mathrm{T}} i\sigma^2\sigma^\mu\partial_\mu\psi_j.$$

In this form, similarly to the $\boxed{1.}$, we see that the transformation of the Weyl fields

$$\psi_j \to M_{ji}\psi_i$$

leads to the global symmetry condition $M^\dagger M = \mathbb{1}$. Therefore, the continuous global symmetry group is $\mathrm{U}(N)$. This transformation can be expressed using the Majorana fields as

$$\Psi_j \to \begin{pmatrix} M_{ji} & 0 \\ 0 & M_{ji}^* \end{pmatrix}\Psi_i.$$

(iii) Using an analogous procedure as in example (ii), with the difference that here the mass term does not mix the left- and right-handed components of the bispinor due to the absence of the gamma matrix which results in terms where both or none of the fields is complexly conjugated, we obtain equations from which we derive the same conditions for the transformation of the Weyl fields as in example (i), i.e., the global symmetry group corresponds to $\mathrm{O}(N)$.

(iv) Similarly to the case of Majorana fields, we can write the Dirac fields in the terms of their Weyl fields

$$\Psi_j = \begin{pmatrix} \psi_j^R \\ \psi_j^L \end{pmatrix}.$$

Using this relation, we rewrite the Lagrangian density

$$\begin{aligned}\mathcal{L} = i\bar{\Psi}_j\gamma^\mu\partial_\mu\Psi_j &= i\begin{pmatrix} \psi_j^{R\dagger} & \psi_j^{L\dagger} \end{pmatrix}\begin{pmatrix} 0 & \sigma^0 \\ \sigma^0 & 0 \end{pmatrix}\begin{pmatrix} 0 & \bar{\sigma}^\mu \\ \sigma^\mu & 0 \end{pmatrix}\partial_\mu\begin{pmatrix} \psi_j^R \\ \psi_j^L \end{pmatrix} \\ &= i\psi_j^{R\dagger}\sigma^\mu\partial_\mu\psi_j^R + i\psi_j^{L\dagger}\bar{\sigma}^\mu\partial_\mu\psi_j^L.\end{aligned}$$

The obtained Lagrangian density corresponds to $2N$ massless Weyl fields, and therefore the continuous global symmetry is $\mathrm{U}(2N)$.

(v) In the case of massive fields, the mass term mixes left- and right-handed components which results in the Lagrangian density

$$\mathcal{L} = i\psi_j^{R\dagger}\sigma^\mu\partial_\mu\psi_j^R + i\psi_j^{L\dagger}\bar{\sigma}^\mu\partial_\mu\psi_j^L - m\psi_j^{R\dagger}\psi_j^L - m\psi_j^{L\dagger}\psi_j^R.$$

Now we can rewrite the ψ_j^L field in terms of a new independent right-handed field χ_j^R:

$$\psi^L = i\sigma^2\chi^{R*}.$$

From now I omit the R notation as both fields are right-handed. The new Lagrangian density is

$$\begin{aligned}\mathcal{L} &= i\psi_j^\dagger \sigma^\mu \partial_\mu \psi_j + i\chi_j^{\mathrm{T}} \sigma^2 \bar{\sigma}^\mu \partial_\mu \sigma^2 \chi_j^* - im\psi_j^\dagger \sigma^2 \chi_j^* + im\chi_j^T \sigma^2 \psi_j \\ &= i\psi_j^\dagger \sigma^\mu \partial_\mu \psi_j + i\chi_j^\dagger \sigma^\mu \partial_\mu \chi_j + im(\psi_j^{\mathrm{T}} \sigma^2 \chi_j)^* + im\chi_j^T \sigma^2 \psi_j,\end{aligned}$$

where i have used the σ^2 transformation of σ^j matrices, hermiticity of σ matrices, the anti-commutativity of the fields and integration per partes together with a vanishing full four-divergence:

$$\begin{aligned}\chi_j^{\mathrm{T}} \sigma^2 \bar{\sigma}^\mu \partial_\mu \sigma^2 \chi_j^* &= -(\chi_j^{\mathrm{T}} \sigma^2 \bar{\sigma}^\mu \partial_\mu \sigma^2 \chi_j^*)^{\mathrm{T}} = -\partial_\mu(\chi_j^\dagger \sigma^{2\mathrm{T}} \bar{\sigma}^{\mu\mathrm{T}} \partial_\mu \sigma^{2\mathrm{T}}) + \chi_j^\dagger \sigma^{2\mathrm{T}} \bar{\sigma}^{\mu\mathrm{T}} \partial_\mu \sigma^{2\mathrm{T}} \chi_j \\ &= \chi_j^\dagger \sigma^2 (\sigma^{0\mathrm{T}} \partial_0 + \sigma^{i\mathrm{T}} \partial_i) \sigma^2 \chi_j = \chi_j^\dagger (\sigma^0 \partial_0 - \sigma^i \partial_i) \chi_j \\ &= \chi_j^\dagger \sigma^\mu \partial_\mu \chi_j\end{aligned}$$

Now we can use the U(2N) symmetry of the kinetic term and make a rotation in the ψ, χ fields by defining new Weyl fields

$$\psi_j = \frac{1}{\sqrt{2}}(\alpha_j - i\beta_j), \qquad \chi_j = \frac{1}{\sqrt{2}}(\alpha_j + i\beta_j).$$

This rotation keeps the kinetic term invariant

$$i\psi_j^\dagger \sigma^\mu \partial_\mu \psi_j + i\chi_j^\dagger \sigma^\mu \partial_\mu \chi_j = i\alpha_j^\dagger \sigma^\mu \partial_\mu \alpha_j + i\beta_j^\dagger \sigma^\mu \partial_\mu \beta_j,$$

which can be proven by a straightforward calculation. For the mass terms we obtain

$$\begin{aligned}\psi_j^{\mathrm{T}} \sigma^2 \chi_j &= \frac{1}{2}(\alpha_j^{\mathrm{T}} - i\beta_j^{\mathrm{T}}) \sigma^2 (\alpha_j + i\beta_j) = \frac{1}{2}\alpha_j^{\mathrm{T}} \sigma^2 \alpha_j + \frac{1}{2}\beta_j^{\mathrm{T}} \sigma^2 \beta_j, \\ \chi_j^T \sigma^2 \psi_j &= \frac{1}{2}(\alpha_j^{\mathrm{T}} + i\beta_j^{\mathrm{T}}) \sigma^2 (\alpha_j - i\beta_j) = \frac{1}{2}\alpha_j^{\mathrm{T}} \sigma^2 \alpha_j + \frac{1}{2}\beta_j^{\mathrm{T}} \sigma^2 \beta_j,\end{aligned}$$

where we have used the fact that

$$\alpha_j^{\mathrm{T}} \sigma^2 \beta_j = -(\alpha_j^{\mathrm{T}} \sigma^2 \beta_j)^{\mathrm{T}} = -\beta_j^{\mathrm{T}} \sigma^{2\mathrm{T}} \alpha_j = \beta_j^{\mathrm{T}} \sigma^2 \alpha_j.$$

Overall, the Lagrangian density is equal to

$$\mathcal{L} = i\alpha_j^\dagger \sigma^\mu \partial_\mu \alpha_j + i\beta_j^\dagger \sigma^\mu \partial_\mu \beta_j + \frac{im}{2}\left[\alpha_j^{\mathrm{T}} \sigma^2 \alpha_j + \beta_j^{\mathrm{T}} \sigma^2 \beta_j + \text{c.c.}\right],$$

which clearly has a separate O(N) continuous global symmetry in both α_j and β_j fields and therefore overall a continous global symmetry O($2N$).

$\boxed{3.}$ We introduce the generators $T^A_{ji} \in \mathfrak{o}(2N)$ of the continuous global symmetry group $\mathrm{O}(2N)$, where the (A) index distinguishes the different generators. The dimension of $\mathrm{O}(2N)$ group, and therefore also the dimension of the $\mathfrak{o}(2N)$ algebra, is $N(2N-1)$. The generators have the following property

$$(T^A)^{\mathrm{T}} = -T^A.$$

These generators act on the spinors – Weyl fields. As the Dirac bispinors are composed of Weyl spinors, the T^A generators act also on the Dirac indices as they mix the left- and right- handed spinors during the rotation needed to transition to the α_j, β_j fields, therefore they are $N(2N-1)$ $2N \times 2N$ matrices or $2 \times 2 \times N \times N$ tensors depending on how we express them.

$$(\Psi_j)_a \to (\Psi_j)^a + i\varepsilon_A T^A_{jiab}(\Psi_i)_b$$

The generators are therefore $N \times N$ complex matrices which act on the *species* index of the Dirac fields.

Using the Noether's theorem we obtain the conserved current

$$\begin{aligned} j^A_\mu &= \frac{\partial \mathcal{L}}{\partial(\partial^\mu(\Psi_j)_a)}(\delta\Psi_j)_a + (\delta\bar{\Psi}_j)_a \frac{\partial \mathcal{L}}{\partial(\partial^\mu(\bar{\Psi}_j)_a)} \\ &= i(\bar{\Psi}_j)_a \gamma_\mu i T^A_{jiab}(\Psi_i)_b + 0 = -(\bar{\Psi}_j)_a \gamma_\mu T^A_{jiab}(\Psi_i)_b \end{aligned}$$